

Project Statement

The Colorado School of Mines senior design team Layered Labs worked on the project “Embedded Electronics in Additively Manufactured Components”, governed by Honeywell. The team conceptually and experimentally verified the possibility of embedded load and thermal sensors in additively manufactured (AM) components. The team developed this system for two test vehicles made of conductive and load bearing materials.

Design Criteria

- 1. Create AM coupons made of structural material and thermally conductive material with embedded sensors that measure load and thermal conditions
- 2. Create AM coupons made of conductive materials with external and internal sensors
- 3. Produce repeatable data with external and internal sensors
- 4. Design test coupon to be geometrically adaptable
- 5. Create a predictive model that correlates measured value from sensor into desired data (applied load / temperature)
- 6. Utilize an AM thermal sensor and prove it properly analyzes material thermal properties
- 7. Utilize a load bearing sensor and prove it measures applied loads

Design & Iterations

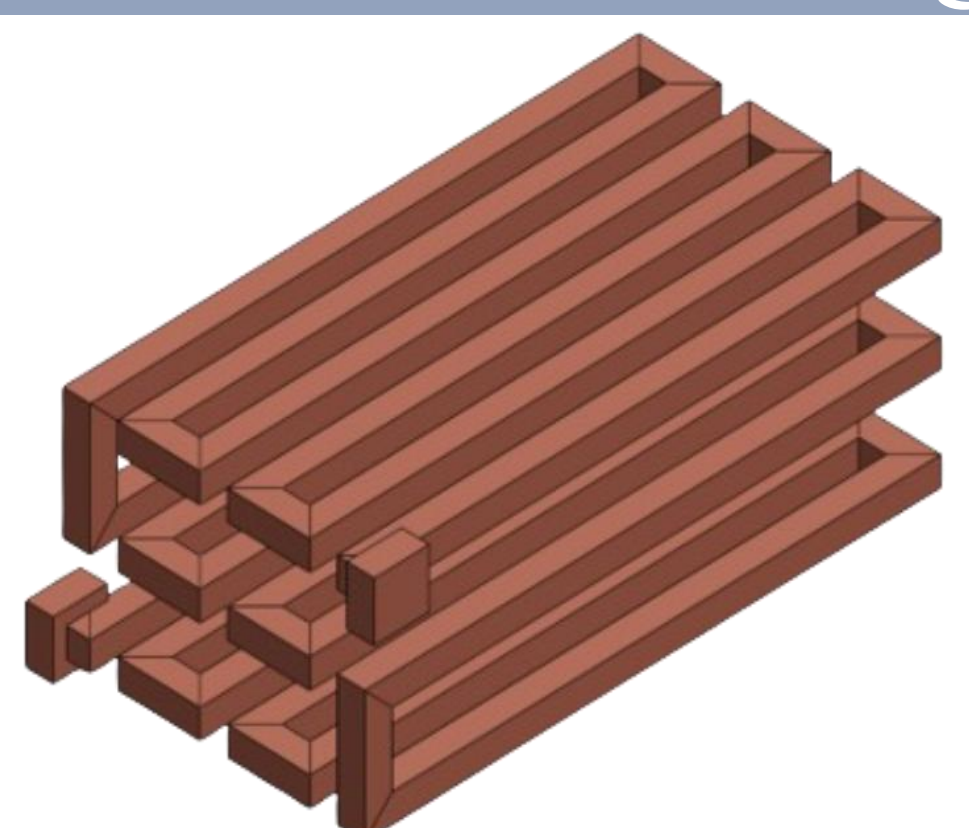


Fig 1. Initial Strain Gauge Design

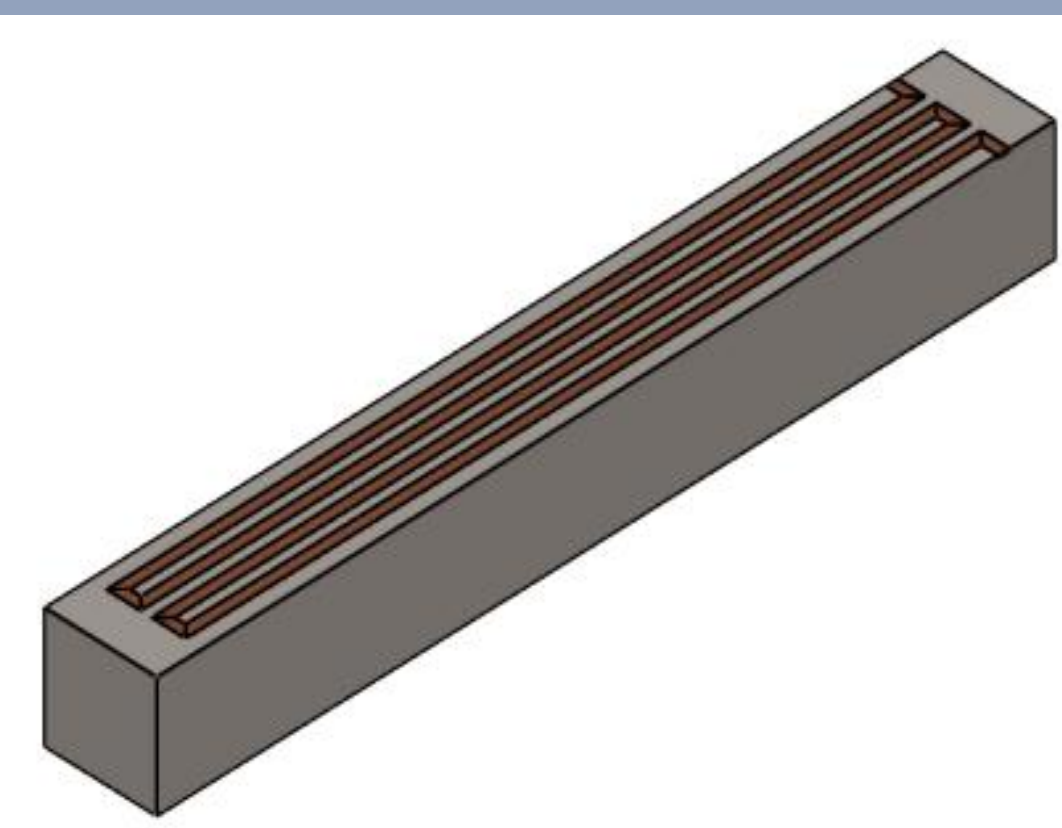


Fig 2. Final Strain Gauge Design

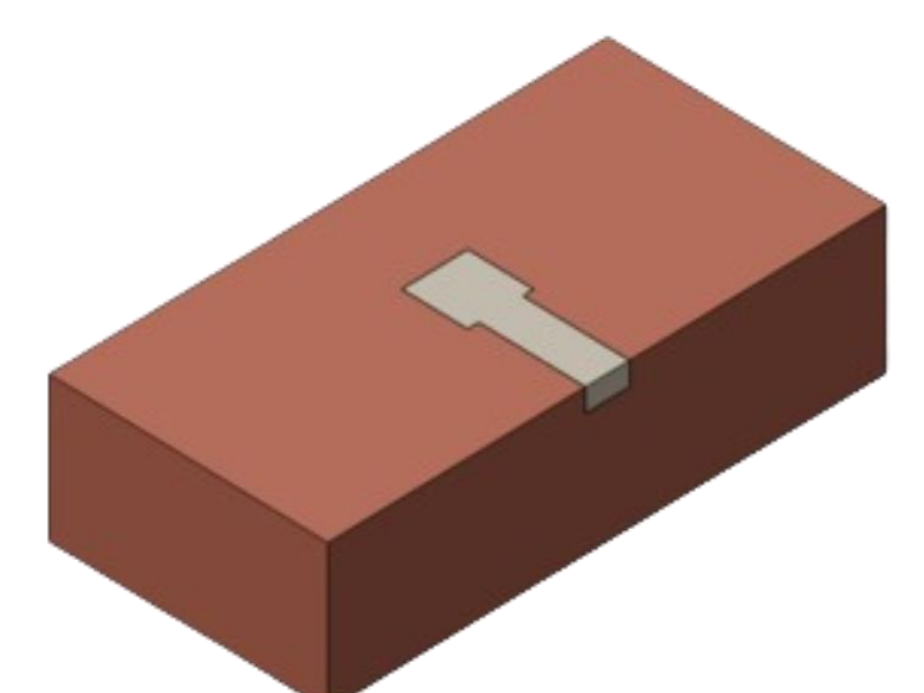


Fig 3. Initial Thermocouple Design

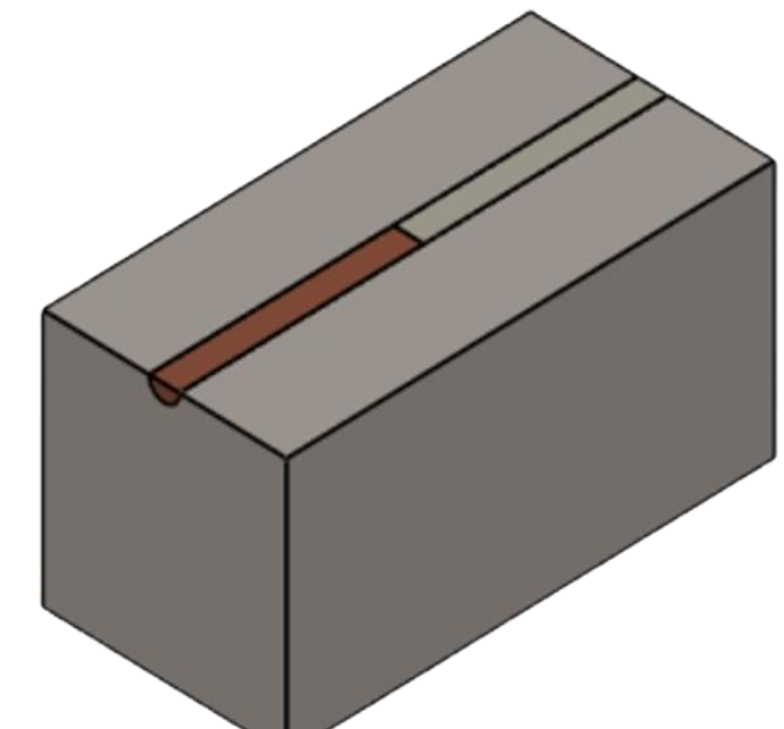


Fig 4. Final Thermocouple Design

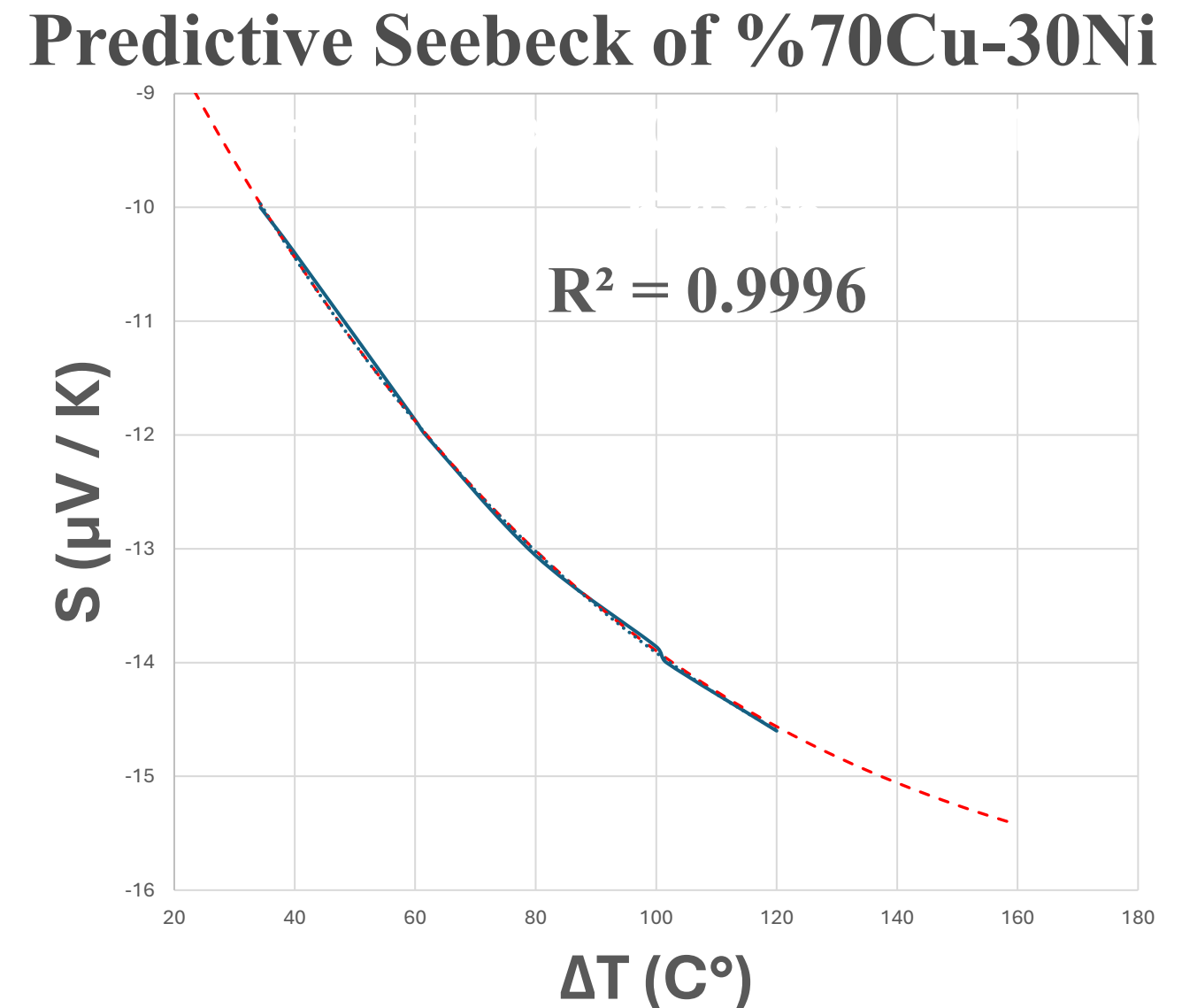
Thermocouple Calculations

$$S = -\frac{\Delta V}{\Delta T} \quad S_{AB} = S_B - S_A = \frac{\Delta V_B}{\Delta T} - \frac{\Delta V_A}{\Delta T}$$
$$S_{tot} = S_{Cu} - S_{CuNi} = 10.157 \frac{\mu V}{K}$$

At $\Delta 20^\circ$

$$\Delta V_{tot} = S_{tot} * \Delta T = 203.132 \mu V$$

Fig 5, 6. Thermocouple Calculations



Strain Gauge Calculations

$$\delta = \delta_{Cu} = \delta_{St}$$
$$P = P_{Cu} + P_{St}$$
$$\epsilon_y = \frac{P}{E_{Cu}A_{Cu} + E_{St}A_{St}}$$
$$\epsilon_x = -\frac{\nu_{Cu}P}{E_{Cu}A_{Cu} + E_{St}A_{St}}$$
$$\Delta R = \frac{\rho_{Cu}NL_i}{w_i^2} \left(\frac{1 + \epsilon_y}{(1 + \epsilon_x)^2} - 1 \right)$$

Fig 7. Strain Gauge Calculations

1000N ~ 0.3454 μΩ

Parameter Development

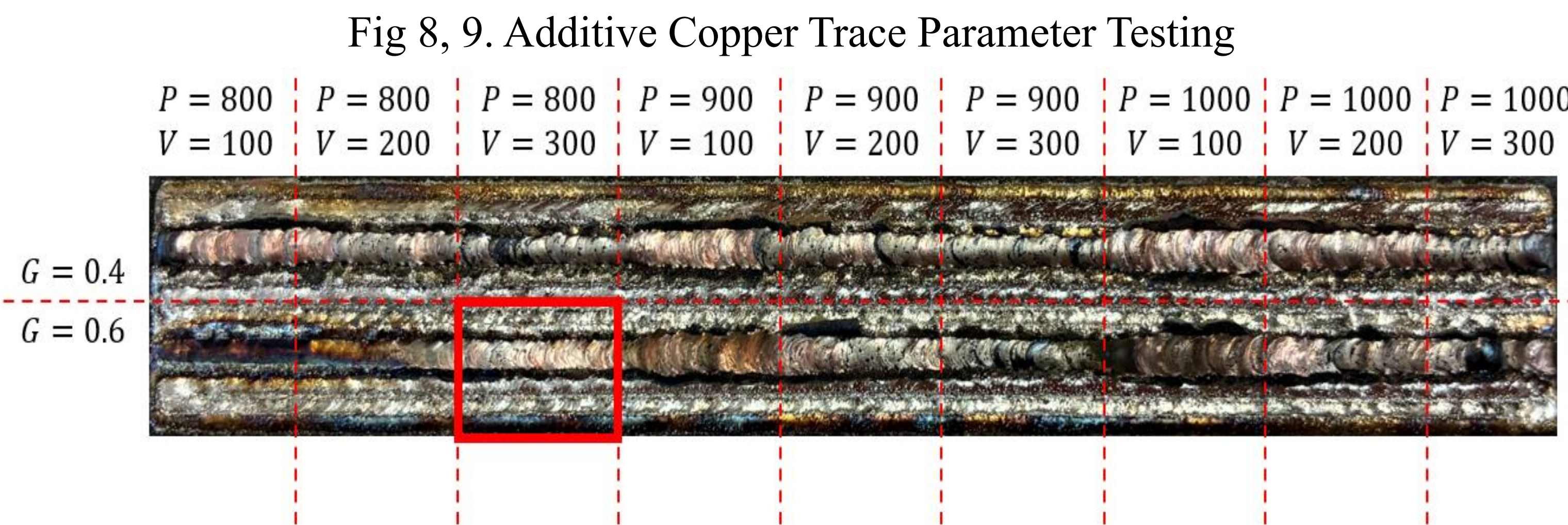
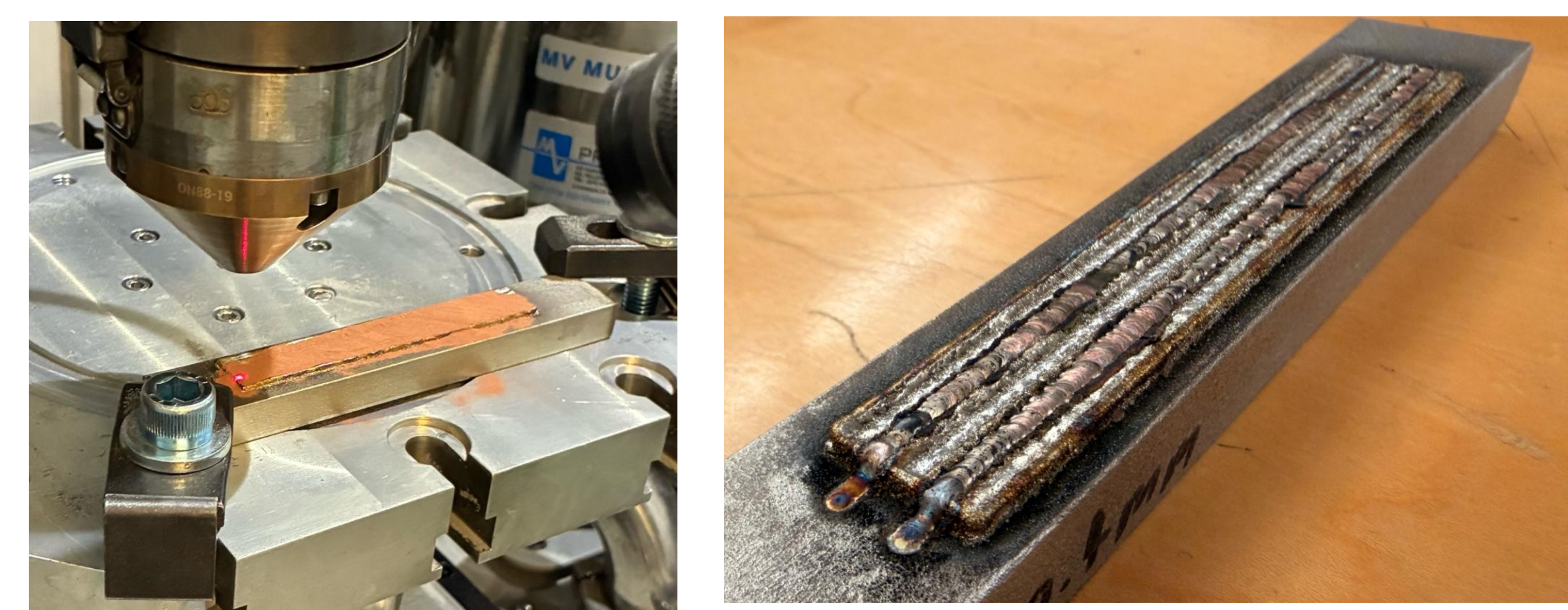


Fig 10. Copper Trace Validation (Ideal Parameters in Red Box – selected through visual inspection and microscopy)

Material Properties

Stress v. Strain AM SS316L Destructive Testing (Axial Extensometer)

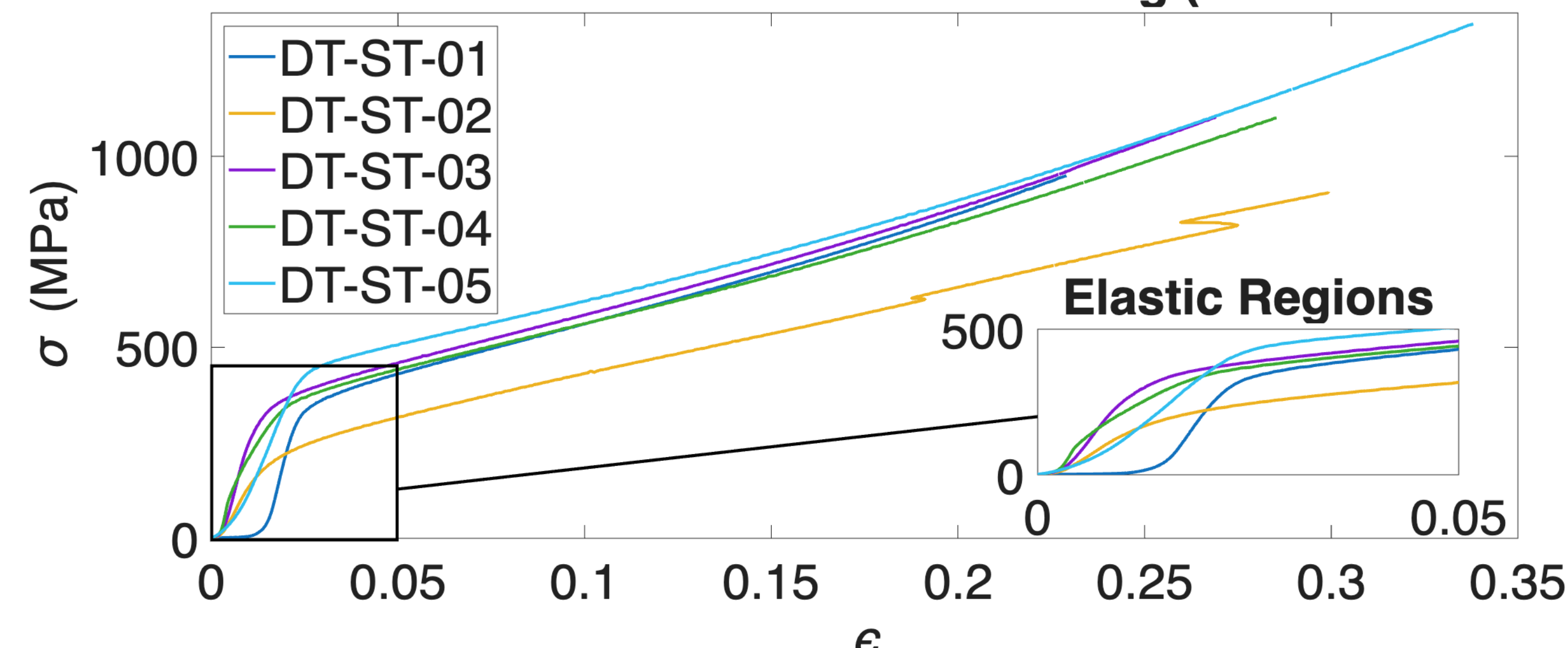


Fig 11. AM SS316L Coupon Material Properties
Table 1. Destructive Material Properties of SS316L

Specimen	E (GPa)	σ _y (MPa)	US (MPa)	Bead Width (mm)	Pool (mm)	Print Orientation
ST-DS-01	39.7	328.2	949.4	1.5	1.5	Vertical
ST-DS-02	18.5	184.5	904.1	0.8	1.3	Vertical
ST-DS-03	37.1	308.7	1102.1	1	1.5	Vertical
ST-DS-04	21	314.5	1100.3	1	1.5	Vertical
ST-DS-05	23.5	428.3	1346.1	1	1.5	Horizontal
Stock 316L	193	170 - 450	1500	-	-	-

Prototype Realization

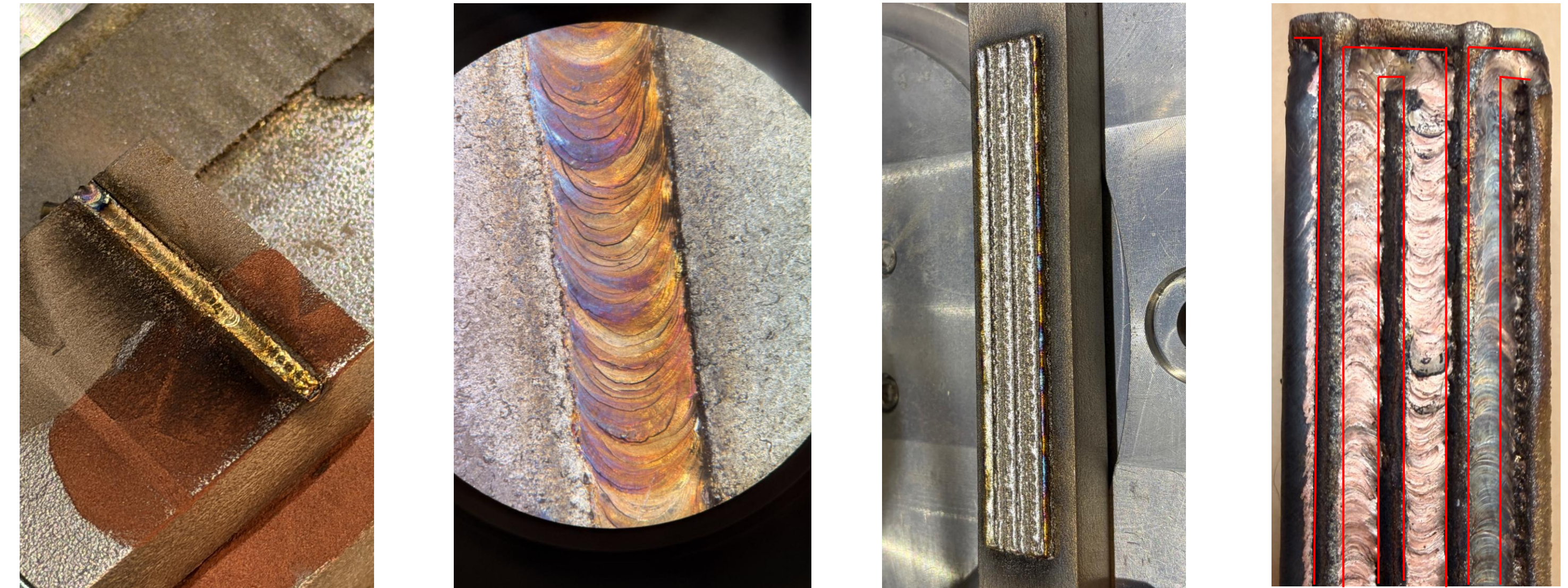


Fig 12, 13. Thermocouple Completed Print Fig 14, 15. Strain Gauge Completed Print

Testing Results

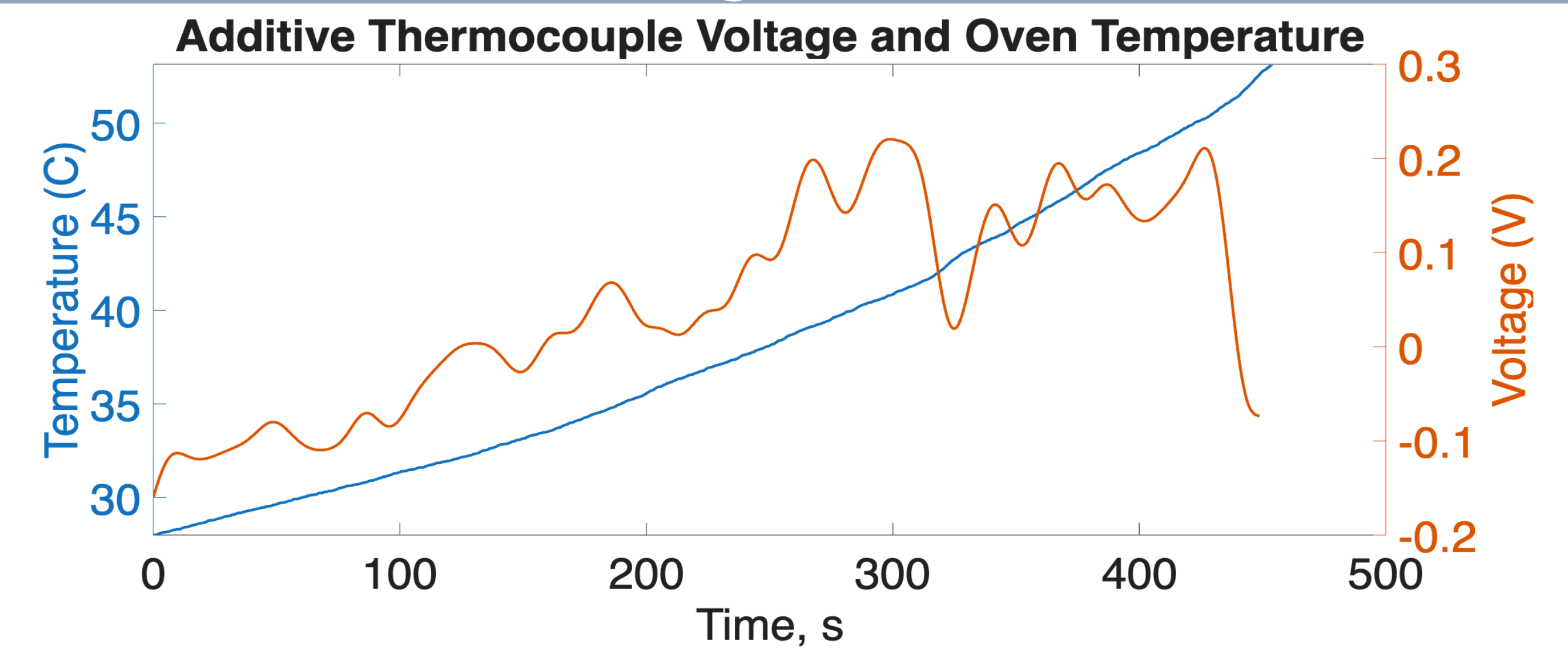


Fig 16. AM Comparison of Thermocouple Voltage to Surface Mounted Thermistor

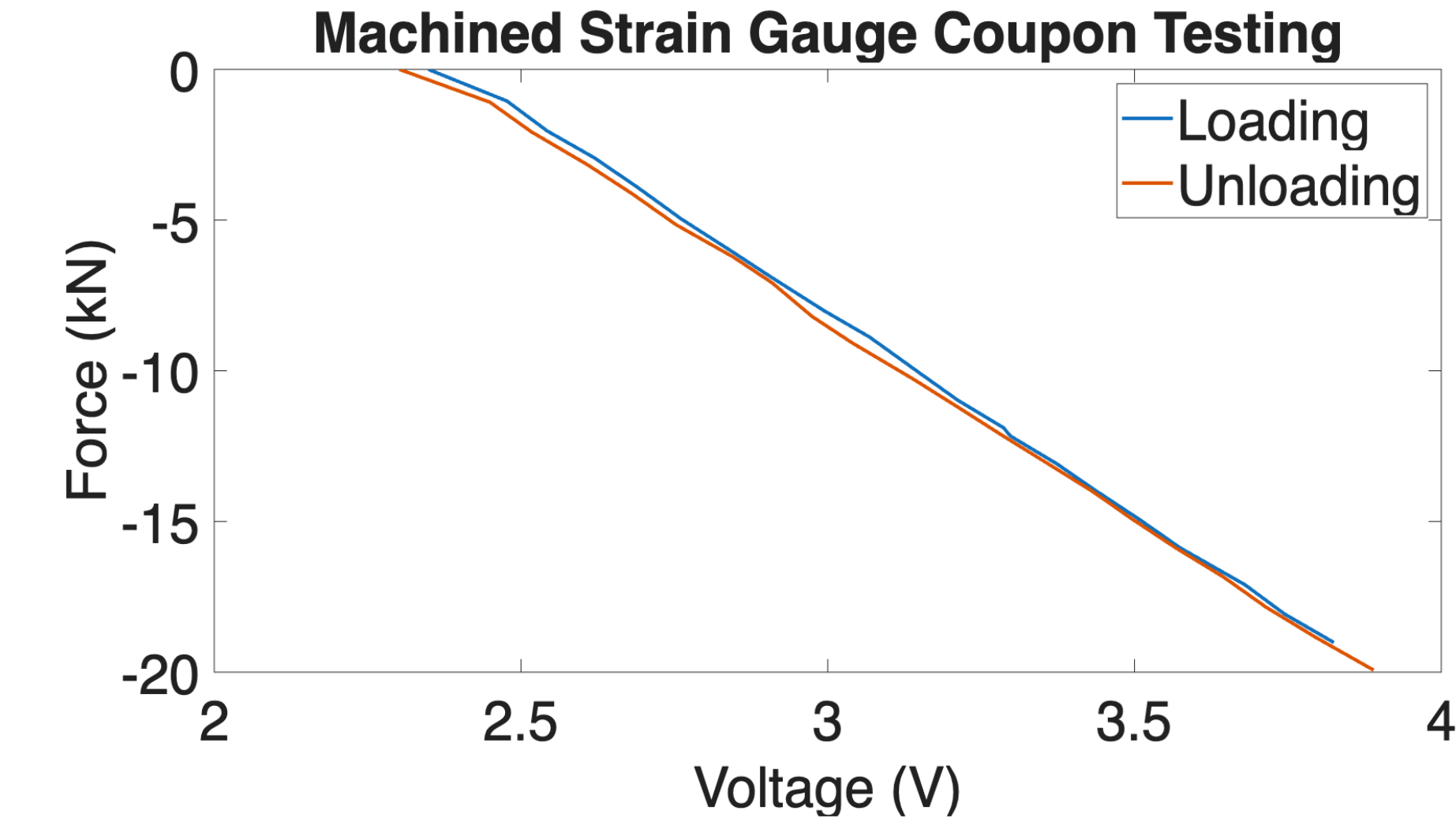


Fig 17. Machined Strain Gauge Load Frame Testing Results

Next Steps

- 1. Source copper alloy and copper-nickel DED powder
- 2. Extensive process parameter development with all powders
- 3. Microscopy and CT scans to check for successful print characteristics
- 4. Create multi-layer test coupons with blown powder (initial concepts)
- 5. Cryogenic temperature testing
- 6. Implement concept into functional components (real environment testing)
- 7. Check consistency of manufacturing process to determine scalability

Acknowledgements

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